MILITARY SPECIFICATION

DISTRIBUTION BOX AND ARBITRARY SCALE METER ASSEMBLY

1. SCOPE

1.1 This specification covers requirements for a distribution box and arbitrary scale meter assembly for use with a 28-volt, direct current vehicular electrical system to supply power to communications equipment mounted in military vehicles. Hereinafter, throughout this specification, the distribution box and arbitrary scale meter assembly shall be referred to collectively as box and meter assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of the invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-E-529	-	Enamel, Alkyd, Semi-Gloss.
TT-P-666	-	Primer Coating, Zinc Yellow, for Aluminum and Magnesium Surfaces.

MILITARY

MIL-R-196	-	Repair Parts for Internal Combustion Engines, Packaging of.
MIL-P-15328	-	Primer (Wash), Pretreatment, Blue (Formula No. 117-B for Metals).
MIL-E-16400	-	Electronic Equipment, Naval Ship and Shore, General Specification.
MIL-G-28923	-	Generator, Engine Accessory: Brushless, Direct Current.

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STANDARDS

FEDERAL

FED-STD-151 - Metal, Test Methods.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking for US Military Property.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810 - Environmental Test Methods.

DRAWINGS

U. S. MARINE CORPS

V1554 - Distribution Box and Voltmeter Tachometer Assembly.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

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3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), prior to submission of any complete production items for acceptance, the contractor shall furnish five first articles for determination of conformance to this specification. Examination and tests shall be those specified herein. Approval of the first articles by the procuring activity shall not relieve the contractor of his obligation to supply box and meter assemblies conforming to this specification. Any changes or deviations of the production items from the first articles shall be subject to the approval of the contracting officer. The first articles shall not be included in the quantity to be delivered to the Government.

3.2 Materials and parts. Unless otherwise specified herein, materials and parts shall conform to the requirements of Drawing No. V1554. Requests for approval of other materials and parts shall be requested in writing to the contracting officer. Approval may be granted where the parts to be used meet the interchangeability requirements of MIL-E-16400.

3.2.1 Screw threads. All screw threads shall conform to H28.

3.3 Design. Design of the box and meter assembly shall be in accordance with Drawing No. V1554 with the exception that cable assembly (Part No. G6283) and instruction plate (Part No. G6261) shall not be supplied with the production units. However, the cable assembly shall be furnished by the contractor for test purposes.

3.3.1 Electrical connections. Electrical interconnection between the distribution box and the meter shall be made by means of the cable assembly (not to be supplied) specified on Drawing No. V1554. Tachometer and voltmeter inputs to the distribution box shall be made through the same cable assembly.

3.4 Construction. The box and meter assembly shall be constructed as specified on Drawing No. V1554.

3.4.1 Ruggedness. The box and meter assembly shall be constructed to withstand both shock and vibration requirements specified herein.

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3.5 Radio frequency (rf) susceptibility. The box and meter assembly shall meet the conducted and radiated susceptibility requirements of MIL-STD-461 for class ID equipment and shall be sufficiently shielded and by-passed so that rf voltages appearing on the interconnecting leads shall not affect meter indication. Interconnecting leads shall be by means of the cable assembly (Part No. G6283) (not to be supplied).

3.6 Reliability. The box and meter assembly shall have a service life of 5,000 hours under all environmental conditions specified herein. The box and meter assembly shall be capable of operating satisfactorily throughout the extended life test specified herein. The mean time between failure shall be not less than 2,000 hours.

3.7 Maintainability. All components and subassemblies, including electronic components within the distribution box, shall be individually replaceable in the field. No loss of watertight integrity shall result from opening the enclosure for field repair or replacement of these components or subassemblies. Use of sealing compounds shall not be required to maintain watertight integrity.

3.8 Performance. The box and meter assembly shall meet the test requirements specified by Drawing No. V1554 under any combination of environmental conditions specified herein.

3.8.1 Environmental conditions. The box and meter assembly shall be capable of normal performance when operated under the following environmental conditions:

(a) Temperature. -40°C to 90°C, operating; -60°C to 90°C nonoperating.

(b) Humidity. 0 to 90 percent relative.

(c) Altitude. Sea level to 15,000 feet with reduction of maximum operating temperature to 71°C for altitudes above 8,000 feet.

(d) Shock. In accordance with MIL-STD-810, method 516.

(e) Vibration. In accordance with MIL-STD-810, method 514, procedure VIII.

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(f) Sea water immersion. Periods of 10 minutes.

(g) Dust atmosphere. In accordance with MIL-STD-810, method 510.

(h) Fungus. In accordance with MIL-STD-810, method 508, procedure 1.

3.9 Dimensions. The box and meter assembly dimensions and mounting provisions shall be as specified on Drawing No. V1554.

3.10 Weight. The weight of the box and meter assembly shall not exceed 2 pounds.

3.11 Color. The color of the box and meter assembly shall match Marine Corps green, No. 23 (see 6.3).

3.12 Finish. All exterior surfaces of the box and meter assembly except the electrical connector shall be painted as follows.

3.12.1 Prime coat. Two coats of primer shall be applied. The first coat of primer shall be in accordance with MIL-P-15328 and the second coat shall be in accordance with TT-P-666.

3.12.2 Finish coat. The finish coat shall be in accordance with class A or B of TT-E-529.

3.13 Marking. Component parts and assemblies shall be marked in accordance with MIL-STD-130.

3.14 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance

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of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Test procedures. Test procedures shall be in accordance with the requirements for each inspection type as specified herein. Testing shall not be performed for record until approval of the test procedure is obtained from the procuring activity.

4.3 Classification of inspections. The box and meter assembly inspection requirements specified herein are classified as follows:

(a) First Article Inspection (see 4.3.1).

(b) Acceptance Inspection (see 4.3.2).

(c) Sample Inspection (see 4.3.3).

4.3.1 First article inspection. The first article inspection shall be conducted on five first article box and meter assemblies which are representative of the production equipment. The first articles shall be numbered one through five and shall retain the same number throughout all first article tests. The first article inspection tests shall consist of all tests listed under "First Article" in Table I. The tests shall be performed in any logical sequence but must be made on the particular numbered unit specified in Table I. All tests shall be witnessed and certified by the cognizant Government inspector.

4.3.2 Acceptance inspection. Acceptance inspection tests specified in Table I shall be performed on each production box and meter assembly submitted for acceptance under the contract. The contractor shall furnish acceptance records in duplicate with each box and meter assembly shipped showing quantitative results for all acceptance inspection tests required by Table I and the contractor shall maintain a record of all acceptance inspection tests. The records shall be available to the cognizant Government inspector.

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TABLE I

INSPECTION

Item	Test	Test Paragraph	Requirement Paragraph	Accept-ance	Sample	First Article				
						1	2	3	4	5
a	Visual Inspection	4.4.1	3.2, 3.3, 3.4, 3.9, 3.11, 3.12, 3.14	X		X	X	X	X	X
b	Electrical Tests	4.4.2	3.8	X		X	X	X	X	X
c	RF Suscep-tibility	4.4.4	3.5		X					X
d	Low Tem-perature	4.4.3.1	3.8.1		X			X	X	
e	High Tem-perature	4.4.3.2	3.8.1		X			X	X	
f	Altitude	4.4.3.3	3.8.1							X
g	Sea Water Immersion	4.4.3.4	3.8.1		X		X	X	X	
h	Sand and Dust	4.4.3.5	3.8.1							X
i	Explosive Atmosphere	4.4.3.6	3.2							X
j	Shock	4.4.3.7	3.8.1		X	X	X	X		
k	Life	4.4.3.8	3.6			X	X	X	X	
l	Fungus	4.4.3.9	3.8.1							X
m	Vibration	4.4.3.10	3.8.1		X	X		X		
n	Humidity	4.4.3.11	3.8.1					X	X	
o	Weight	4.4.5	3.10		X	X				

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4.3.3 Sample inspection. The sample inspection shall consist of the tests specified in Table I on the sampling plan test quantity specified in 4.3.3.1. Box and meter assemblies submitted for test shall be selected at random by the cognizant Government inspector and shall have passed the acceptance inspection tests.

4.3.3.1 Sampling plan. Sampling plan quantity shall be selected as follows:

Consecutively Produced Assemblies	Test Quantity
1 - 25	1
26 - 200	1
201 - 500	1
501 - 1000	1

4.3.3.2 Rejection and retest. A failure of a box and meter assembly during the sample inspection tests shall be handled as follows.

4.3.3.2.1 Random type failure. If the box and meter assembly fails any of the sample inspection tests and the cause of failure is determined to be of the random type, the failed component shall be replaced and the test program continued.

4.3.3.2.2 Other failures. If the box and meter assembly fails any of the sample inspection tests and the cause of failure is determined to be other than random type, corrective action shall be taken on the test assembly and on each assembly in the lot represented by the test assembly. At the discretion of the cognizant Government inspector, one of the following programs shall be followed:

(a) The test which was being performed when the failure occurred shall be repeated on the corrected test assembly and the remainder of the test program completed.

(b) The test which was being performed when the failure occurred and any previous tests performed which may have contributed to the failure shall be repeated

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on the corrected test assembly and the remainder of the test program completed.

(c) The sample inspection shall be repeated on another assembly which has received the corrective action.

4.4 Inspection and test. Unless otherwise specified herein, each test in this section shall be conducted under normal ambient conditions. Normal ambient conditions shall be a temperature of 25° ±10°C, an altitude of 0 to 500 feet and a relative humidity of 0 to 90 percent. During tests where operation is specified, the box and meter assembly shall be connected in accordance with Figure I. A direct current (dc) voltage of 28 Vdc ±1 percent and an alternating current (ac) voltage of 21.1 volts root mean square ±2 percent shall be applied to the assembly in accordance with Figure I. Frequency of the ac voltage shall be between 100 hertz (Hz) and 150 Hz. The mode selector switch may be operated in either voltage or frequency mode.

4.4.1 Visual inspection. The box and meter assembly shall be visually inspected to determine correctness of materials, construction, color, finishes, marking and identification, workmanship, completeness, component fits in their respective positions, connections and general appearance.

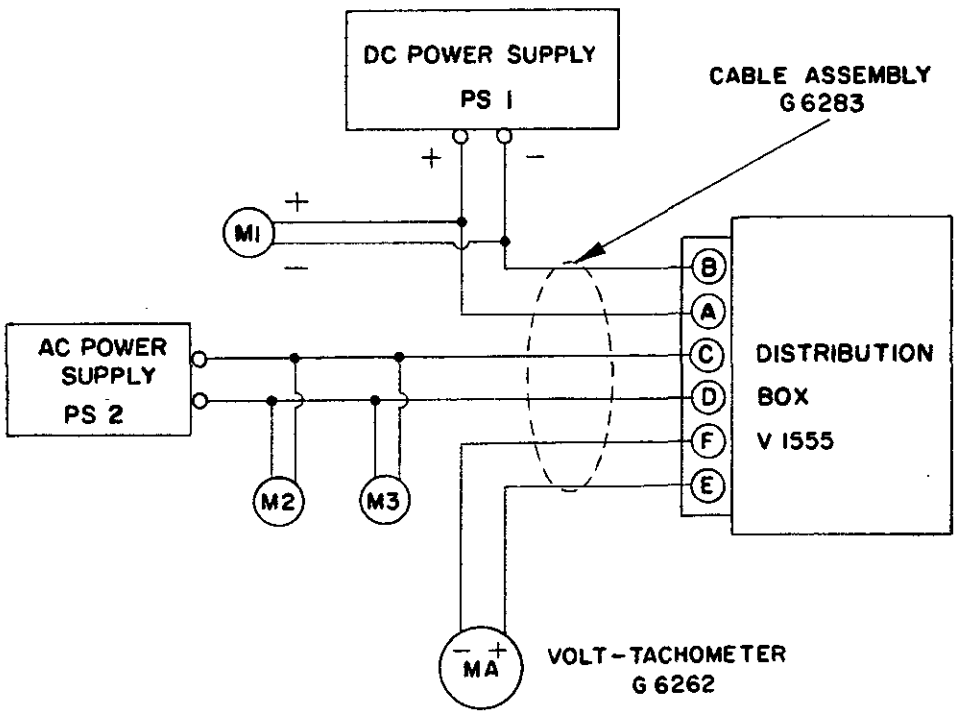
4.4.2 Electrical tests. The box and meter assembly shall be tested in accordance with electrical test requirements of Drawing No. V1554.

4.4.2.1 Test equipment requirements. Equipment required for test purposes shall be as specified by Drawing No. V1554. Cable assembly, Part No. G6283, shall be furnished by the contractor for test purposes only.

4.4.3 Service condition tests. The box and meter assembly shall be visually inspected, both internally and externally, immediately before and after each of the following service condition tests. Any signs of leakage, corrosion, damaged parts or other defects shall be cause for rejection of the test assembly.

4.4.3.1 Low temperature. The box and meter assembly shall be tested in accordance with MIL-STD-810, method 502, procedure I. The assembly shall be operated during the test (see 4.4).

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NOTE: FOR IDENTIFICATION OF COMPONENTS REFERENCED SEE DRAWING NO. V1554

FIGURE I. DIAGRAM FOR TEST CONNECTION OF BOX AND METER ASSEMBLY

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4.4.3.1.1 Test data. The assembly shall be subjected to the tests of 4.4.2.

4.4.3.1.2 Storage temperature. Storage temperature shall be -60°C for a period of 4 hours.

4.4.3.1.3 Operating temperature. Operating temperature shall be -40°C for a period of 8 hours.

4.4.3.2 High temperature. The box and meter assembly shall be operated in accordance with 4.4 for 96 hours in an ambient temperature of 90°C. There shall be no damage to the assembly as a result of the test and the unit shall meet the test requirements of 4.4.2 at the end of each 16 hours during the test.

4.4.3.3 Altitude. The box and meter assembly shall be operated at a pressure altitude of 15,000 feet at an ambient temperature of 71°C for a 1-hour period. The assembly shall be operated during the test (see 4.4). The tests of 4.4.2 shall be conducted at the end of each 15 minutes during the test.

4.4.3.4 Sea water immersion. This test shall be performed using a sea water solution prepared in accordance with FED-STD-151, method 812. During the test, sand and dust shall be released into the top of the test tank at a uniform rate. The sand and dust specified in MIL-STD-810, method 510, shall compose 75 percent of the total and the remainder of the material shall be sand which will pass through a 40-mesh screen but will not pass through a 100-mesh screen. The sand and dust shall be released at a rate of 8 ounces per minute per square foot of water surface. The distribution box shall be disassembled and reassembled 10 times prior to testing.

4.4.3.4.1 Performance of test. While operating under conditions specified in 4.4, the assembly shall be gradually immersed to a depth of 3 feet in sea water and gradually removed. The change of water depth from 0 to 3 feet shall extend for 5 minutes with a total cycle time of 10 minutes required to bring the assembly back to 0 feet. Immersion of the assembly shall consist of 3 complete cycles. Upon completion of test, the assembly shall meet test requirements in 4.4.2.

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4.4.3.5 Sand and dust. The box and meter assembly shall be subjected to sand and dust test in accordance with MIL-STD-810, method 510, procedure I. The assembly shall be operated as specified in 4.4 during the test. The assembly shall be subjected to the tests in 4.4.2 as specified by MIL-STD-810.

4.4.3.6 Explosive atmosphere. The box and meter assembly shall be subjected to an explosive atmosphere test in accordance with MIL-STD-810, method 511, procedure I.

4.4.3.7 Shock.

4.4.3.7.1 Basic design shock. The box and meter assembly shall be subjected to basic design shock test in accordance with MIL-STD-810, method 516, procedure I, with sawtooth pulse having a peak value of 20 g's and 11-millisecond duration. Upon completion of the test, the assembly shall meet test requirements in 4.4.2.

4.4.3.7.2 Bench handling shock. The box and meter assembly shall be subjected to the bench handling shock test in accordance with MIL-STD-810, method 516, procedure V. At the completion of test, the assembly shall meet test requirements in 4.4.2.

4.4.3.8 Life. The box and meter assembly shall be operated as specified in 4.4 for 5,000 hours at room ambient conditions. Repair after every 2,000 hours of operation is permissible. Repair shall be limited to cleaning, minor touch up of painted or insulated surfaces and tightening of mechanical fasteners. The assembly shall be tested at the end of each 100 hours in accordance with 4.4.2.

4.4.3.9 Fungus. The box and meter assembly shall be subjected to a fungus test in accordance with MIL-STD-810, method 508, procedure I. At the completion of the fungus test, the assembly shall meet the test requirements in 4.4.2.

4.4.3.10 Vibration. The box and meter assembly shall be subjected to vibration tests in accordance with MIL-STD-810, method 514, procedure VIII. The assembly shall be operated as specified in 4.4 during the test. At the completion of the test, the assembly shall meet test requirements in 4.4.2.

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4.4.3.11 Humidity. The box and meter assembly shall be subjected to humidity test in accordance with MIL-STD-810, method 507, procedure II. The assembly shall be operated as specified in 4.4 during the test. Tests in 4.4.2 shall be conducted as specified by MIL-STD-810, method 507, procedure II and during the 12th, 32nd, and 40th hour of each 48-hour cycle.

4.4.4 Radio frequency interference. The box and meter assembly shall be subjected to the radiated susceptibility tests of MIL-STD-462, methods RS01, RS02, RS03, and RS04 and to the conducted susceptibility tests of MIL-STD-462, methods CS01 and CS02.

4.4.5 Weight test. The box and meter assembly shall be weighed on a calibrated scale. The weight shall not exceed 2 pounds.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging of the box and meter assembly shall be level A, B or C in accordance with MIL-R-196, as specified (see 6.2). The intermediate packaging quantity shall be 20 for levels A and B.

5.2 Packing. Packing of the box and meter assembly shall be level A, B, or C in accordance with MIL-R-196, as specified (see 6.2).

5.3 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The identification shall be composed of the following information listed in the order shown:

Distribution Box and Arbitrary Scale Meter Assembly

Military Part Number

Federal Stock number or other identification number as specified in the procurement document.

Manufacturer's Part Number

Manufacturer's Serial Number

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Contract or Order Number

Date of Manufacture

Manufacturer's Name or Trademark

Name of Contractor (if different from manufacturer)

6. NOTES

6.1 Intended use. The box and meter assembly covered by this specification is intended for use in military vehicles and military landing craft to provide for monitoring output voltage and generator speed of dc generators manufactured in accordance with MIL-G-28923.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Whether first article is required (see 3.1).

(c) Military part number (see 5.3).

(d) Level of preservation, packaging and packing (see 5.1 and 5.2)

6.3 Color. Color chips for Marine Corps green, No. 23 may be obtained from Commandant of the Marine Corps, Headquarters U. S. Marine Corps (CSY-10), Washington, D. C., 20380.

6.4 First article. The first article shall be examined and tested for approval at the contractor's plant or at an independent commercial testing laboratory acceptable to the procuring activity. First article tests shall be witnessed by a representative of the procuring activity. Approval of the first article shall be by the contracting officer.

Preparing activity:
Navy - MC
Project No. 6110-N061

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